

For $\mathbb{R}^3$ it is convenient to introduce notation for the standard basis

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

- (1) (a) At what point does the curve $t \mapsto t\mathbf{i} + (2t - t^2)\mathbf{k}$ intersect the paraboloid $z = x^2 + y^2$?
 (b) Find centre and radius of the sphere $x^2 + y^2 + z^2 - 8x + 2y + 6z + 1 = 0$.

- (2) (a) Find a parametrization of the curve of intersection of the paraboloid $z = 4x^2 + y^2$ and the parabolic cylinder $y = x^2$.
 (b) Find a parametrization of the tangent line to the curve $F(t) = (t^2 - 1)\mathbf{i} + (t^2 + 1)\mathbf{j} + (t + 1)\mathbf{k}$ at the point $-\mathbf{i} + \mathbf{j} + \mathbf{k}$.

- (3) Suppose the trajectories of two moving particles are given by the curves $\gamma_1(t) = t^2\mathbf{i} + (7t - 12)\mathbf{j} + t^2\mathbf{k}$ and $\gamma_2(t) = (4t - 3)\mathbf{i} + t^2\mathbf{j} + (5t - 6)\mathbf{k}$ for $t \geq 0$. Will the particles collide?

- (4) (a) Find the point of intersection of the line $F(t) = (2 + 3t)\mathbf{i} - 4t\mathbf{j} + (5 + t)\mathbf{k}$ and the plane $4x + 5y - 2z = 18$.
 (b) Find the equation of the plane that contains the three points $\mathbf{i}$, $2\mathbf{j}$ and $3\mathbf{k}$.
 (c) Show that the planes $x + y - z = 1$ and $2x - 3y + 4z = 5$ are neither parallel nor perpendicular. Compute the angle between these planes.
 (d) Find the equation of the plane through the point $2\mathbf{i} + \mathbf{j}$ parallel to the plane $x + 4y - 3z = 1$.

- (5) (a) The position of a moving particle at time $t \geq 0$ is given by the map $F(t) = t^2\mathbf{i} + 5t\mathbf{j} + (t^2 - 16t)\mathbf{k}$. At what time is the speed the least?
 (b) Show that if a particle moves at constant speed, then velocity and acceleration vectors are perpendicular.

- (6) Find a parametrization of the lower half of the ellipsoid with half-axes $(a, b, c) = (2, 1, 3)$.

- (7) Find a parametrization of the surface of revolution obtained by rotating the curve $x = 4y^2 - y^4$, $-2 \leq y \leq 2$ about the y-axis. Sketch the resulting surface.

- (8) Do the following limits exist?

$$\begin{aligned} \text{(i)} \quad & \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4} \\ \text{(ii)} \quad & \lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} \end{aligned}$$

- (9) Use the definition of partial derivatives as limits to find f_x and f_y for

$$\begin{aligned} \text{(i)} \quad & f(x, y) = xy^2 - x^2y \\ \text{(ii)} \quad & f(x, y) = \frac{x}{x + y^2} \end{aligned}$$

(10) Use implicit differentiation to find $\partial z/\partial x$ and $\partial z/\partial y$ for

- (i) $yz = \ln(x + z)$
- (ii) $x - z = \arctan(yz)$.

(11) Verify Clairaut's Theorem for the following functions.

- (i) $f(x, y) = x \sin(x + 2y)$
- (ii) $f(x, y) = x^4 y^2 - 2xy^5$.

(12) Find all second partial derivatives.

- (i) $f(x, y) = \ln(3x + 5y)$
- (ii) $z = \frac{x}{x+y}$.

(13) Find the indicated partial derivatives.

- (i) $f(x, t) = x^2 e^{-ct}; f_{ttt}, f_{txx}$
- (ii) $f(x, y, z) = \cos(4x + 3y + 2z); f_{xyz}, f_{zyx}$.

(14) The ellipsoid $4x^2 + 2y^2 + z^2 = 16$ intersects the plane $y = 2$ in an ellipse. Find the equation of the tangent line to the ellipse at the point $(1, 2, 2)$.